Faculty Timetable Generator Using C

C Mini Project

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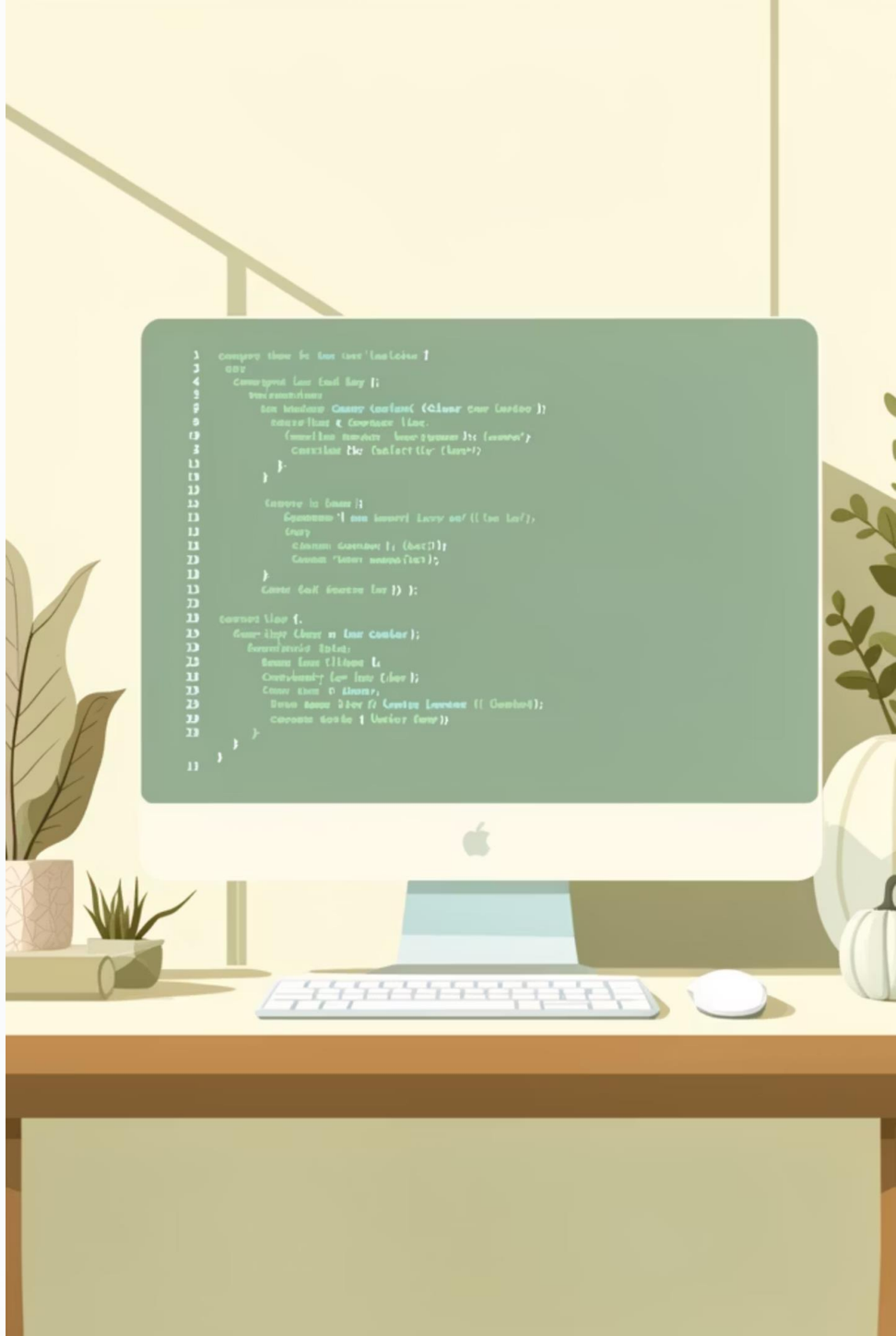
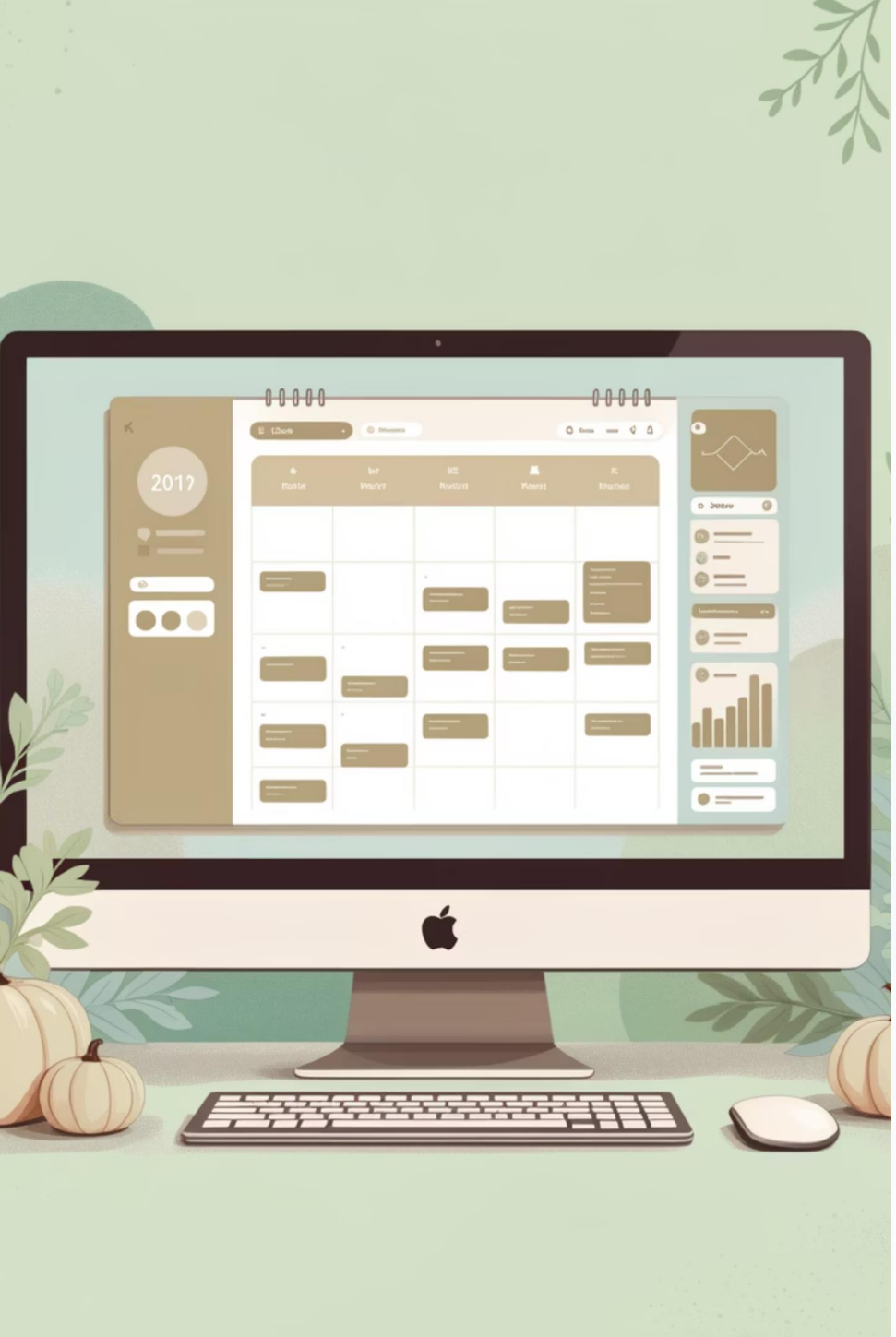




Introduction

Timetable management is a critical task in educational institutions.

In many colleges, timetable scheduling is:

- Done manually
- Time-consuming
- Prone to human errors
- Difficult to modify

Common Problems:

- Faculty assigned to two classes at same time
- Batch assigned multiple subjects in same period
- No centralized data storage

To solve this, we developed a **Faculty Timetable Generator using C language**.

Problem Statement

Educational institutions require an automated system that prevents scheduling conflicts and manages faculty timetables efficiently. Manual scheduling leads to double booking, conflicts, and poor time management.

1

Prevents Faculty Clashes

Ensures no double booking of instructors

2

Prevents Batch Clashes

Ensures no overlapping batch assignments

3

Persistent Storage

Stores timetable data permanently in CSV files

4

Real-Time Updates

Displays slot availability and allows easy updates

Project Objectives

01

Develop Timetable System

Build using C programming language with modular approach

02

Store Data in CSV

Maintain timetable data in comma-separated value files

03

Prevent Conflicts

Implement automated conflict-checking mechanism

04

Slot Status Tracking

Provide FREE, OCCUPIED, and BLOCKED status indicators

05

Separate File Management

Maintain distinct files for faculty and batches

06

Modular Design

Use organized, scalable code structure

Technologies & Concepts Used

Development Tools

- GCC Compiler
- Code Editor (VS Code / Dev C++)

Core Concepts

- Structures
- 2D Arrays
- File Handling (CSV)
- Header Files
- Modular Programming
- String Handling
- Conditional Statements
- Loops

Programming Language

- C Programming

System Architecture

The project is divided into multiple modules for improved readability, maintainability, and scalability.

main.c

Handles program execution and user interaction

fileio.c

Manages loading, saving, and file existence checking

timetable.c

Handles slot checking, updating, and initialization

ui.c

Manages display, input, and batch selection

Data Structures Used

BatchSlot Structure

```
typedef struct {
    char faculty[50];
} BatchSlot;
```

Stores faculty name assigned to batch slot

FacultySlot Structure

```
typedef struct {
    char batch[50];
    char subject[50];
} FacultySlot;
```

Stores batch and subject names

Timetable Dimensions

5 Days (N = 5)

4 Periods per Day (M = 4)

Total Slots = 20

```
BatchSlot batch[N][M];
FacultySlot faculty[N][M];
```

User Interface

The system displays slot view:

Example Output:

```
Slot 0 (Day 0, Period 0): FREE
Slot 1 (Day 0, Period 1): OCCUPIED
Slot 2 (Day 0, Period 2): BLOCKED
```

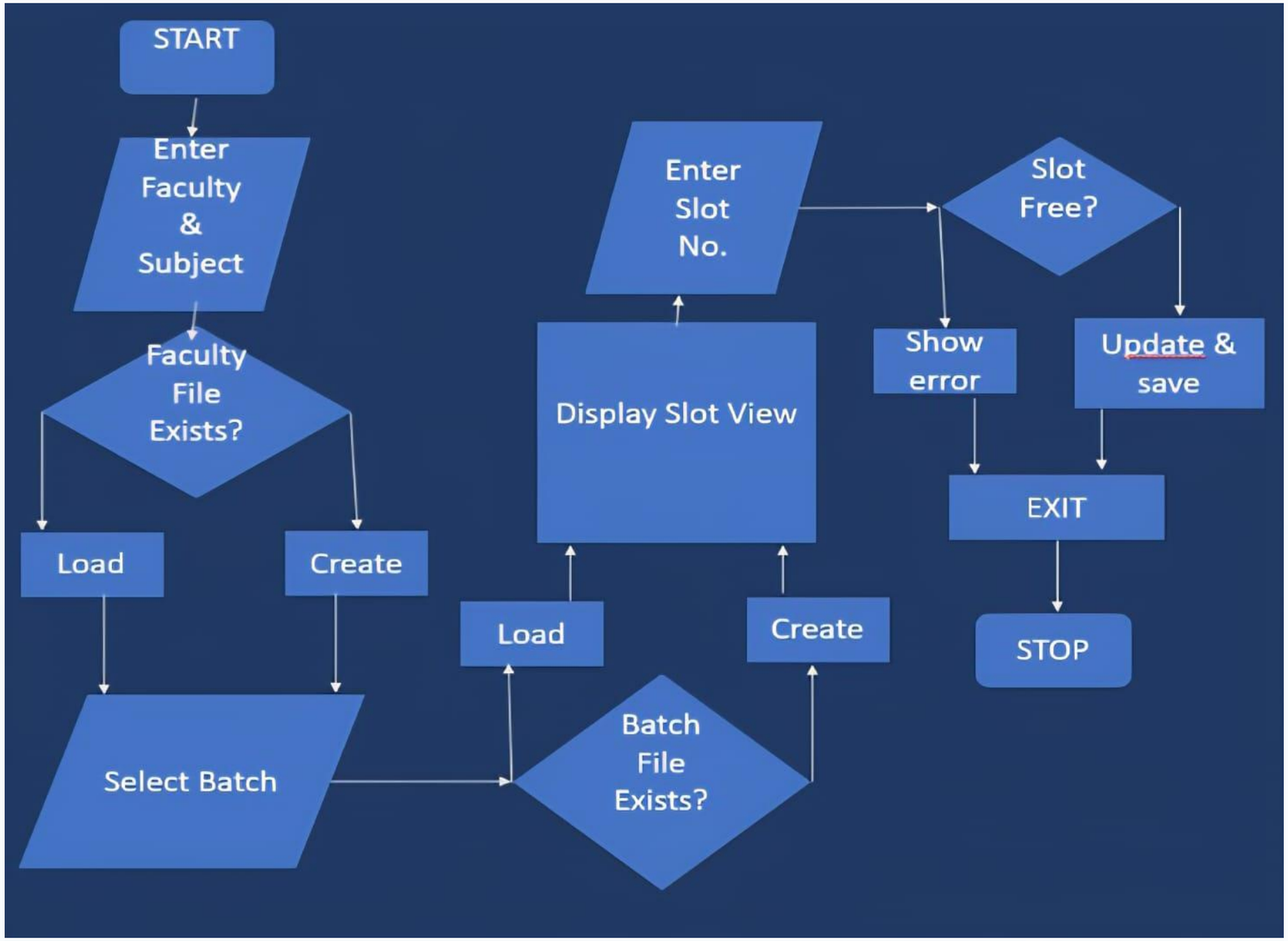
Legend:

- FREE → Can assign
- OCCUPIED → Batch already taken
- BLOCKED → Faculty already busy

User can:

- Select slot number
- Exit selection
- Choose different batch

Workflow & Flowchart



Workflow

1. Start program
2. Enter faculty name and subject
3. Load or create faculty timetable file
4. Select batch
5. Load or create batch timetable file
6. Display slot status (FREE / OCCUPIED / BLOCKED)
7. Select slot
8. Check if slot is free (both faculty & batch)
9. If free → update and save files
10. Repeat or exit

File Handling Mechanism

Each timetable is stored in CSV format for persistent storage, easy readability, and external editing.

Faculty File Example

```
day,period,batch,subject
0,0,EMPTY,EMPTY
1,0,CS101,Programming
1,1,CS202,Database
2,0,CS101,Programming
```

Batch File Example

```
day,period,faculty
0,0,EMPTY
1,0,Dr. Smith
1,1,Prof. Johnson
2,0,Dr. Smith
```

file_exists()

Checks if timetable file exists

load_batch()

Loads batch timetable data

save_batch()

Saves batch timetable to file

load_faculty()

Loads faculty timetable data

save_faculty()

Saves faculty timetable to file

Advantages

Conflict Prevention

Successfully prevents scheduling conflicts

Lightweight

Simple and efficient system

Modular Code

Organized, maintainable structure

Easy Management

Simple file management

Scalable

Suitable for small institutions

Extensible

Easy to modify and enhance

Conclusion

The Faculty Timetable Generator:

- Successfully prevents scheduling conflicts
- Stores data efficiently using CSV files

Demonstrates core C concepts:

- Structures
- File Handling
- Modular Programming
- 2D Arrays
- String Handling

THANK YOU